Listing 11.10 describes a possible implementation of such a simulation leveraging an S-tree to perform range-based queries and filter, for each particle, the surrounding particles with which it is relevant to compute the interaction. The catch is that the k-d tree (or equivalent DS) used needs to be updated after every step of the simulation (since the position of each particle changes), but still, even so, the speedup that can be obtained is impressive.

11.4.3 Multidimensional DB queries optimization

As we have seen in chapter 9, k-d trees support multidimensional range queries (MDRQ), searches that select intervals in two or more dimensions of a multidimensional search space (for instance, as we suggested in chapter 9, a query that searches every employee between thirty and sixty years old and that earns between 40 and 100 thousand dollars per year).